

AI 4403 : Basic Financial Instruments

Easwar Subramanian

TCS Innovation Labs, Hyderabad

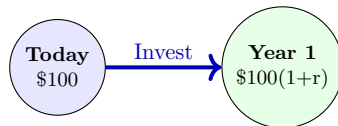
Email : easwar.subramaniam@ai.iith.ac.in

August 6, 2026

- 1 Fixed Deposits
- 2 Bonds
- 3 Forwards and Futures
- 4 Options

Fixed Deposits

- ▶ Would you prefer **\$100 today** or **\$100 one year from now**?
- ▶ **Why standard preference favors today:**
 - ★ Opportunity Cost (Earning potential)
 - ★ Inflation (Purchasing power risk)
 - ★ Default / Uncertainty risk
- ▶ Money has a **temporal cost**: r (Interest Rate).

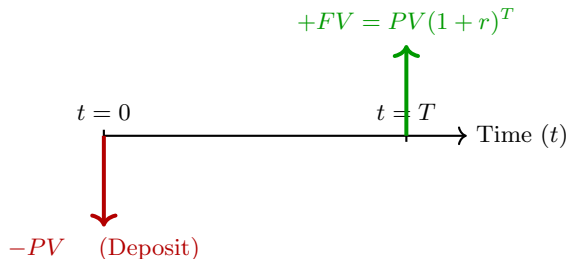


Fundamental Valuation Principles

- ▶ **Compounding (Future Value):** $FV = PV \cdot (1 + r)^T$
- ▶ **Discounting (Present Value):** $PV = \frac{FV}{(1+r)^T}$

Fixed Deposits (FD) – The Baseline Instrument

- **Definition:** Deposit a single sum PV today in exchange for a **guaranteed** payment FV at maturity T .
- **Cash Flow Structure:** Exactly *one* outflow now, *one* inflow later.



Key Takeaway

In an FD, future cash flows are **deterministic** and single-period. Pricing is simply matching discounting with the guaranteed rate r .

Bonds

What is a Bond? A standardized debt instrument where an entity borrows money directly from investors for a defined period at a fixed interest rate.

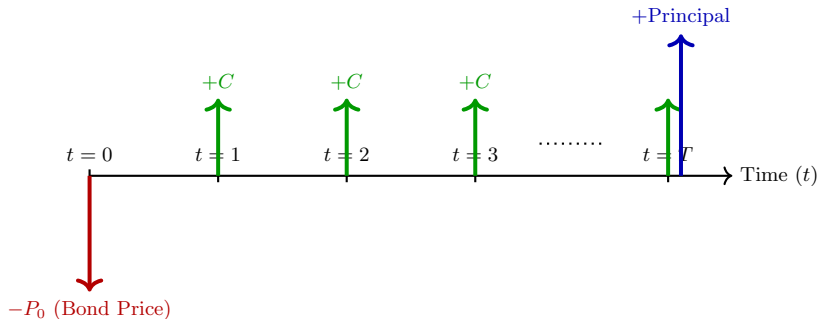
Why do Companies Issue Bonds?

- ▶ **Bypassing the Bank:** Bank loans can be restrictive and carry higher interest rates.
- ▶ **Scale:** Large projects (infrastructure, expansion) require more capital than a single bank might risk.
- ▶ **Flexibility:** Companies can structure maturities and covenants to suit their specific cash flow needs.

Why do Investors Buy Them?

- ▶ **Higher Yields:** Often provide better returns than standard bank FDs.
- ▶ **Regular Income:** Periodic interest payments (coupons) provide steady cash flow.
- ▶ **Tradability:** Unlike FDs, bonds can be bought and sold on the open market *before* maturity.

- **Cash Flow Structure:** Moving from one single payment (FD) to a *sequence* of periodic payments (Coupons, C) plus the Principal return at maturity T .
- Let's visualize a standard Annual Coupon Bond:

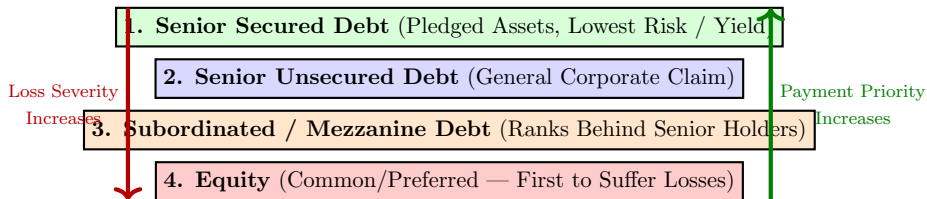


Pricing by Discounting a Sequence

Because the cash flows are deterministic, we simply apply the Present Value formula to *each* cash flow and sum them up:

$$P_0 = \frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} + \cdots + \frac{C + \text{Principal}}{(1+r)^T} = \sum_{t=1}^T \frac{C_t}{(1+r)^t}$$

- ▶ **Priority of Claims:** In financial distress, cash flows strictly flow top-to-bottom.
- ▶ Higher priority = lower default risk = lower yield required by investors.



The Rating Impact

Rating agencies (CRISIL, Moody's) assign distinct credit ratings down this stack for the *same* issuing entity.

The Key Difference: Tradability & Price Risk

An FD is a private contract with a bank; its value doesn't fluctuate. A Bond is a **tradable security**. If market interest rates change, the market price of an existing bond will fluctuate!

- ▶ Bonds are born in the **Primary Market** (issued by companies to initial investors).
- ▶ However, they spend most of their lives in the **Secondary Market**.

Why would an investor trade a bond before maturity (T)?

- ▶ **Liquidity Needs:** The investor suddenly needs cash today and cannot wait years for the principal repayment.
- ▶ **Speculation:** The investor anticipates changes in market interest rates and wants to capture capital gains.
- ▶ **Portfolio Rebalancing:** Institutional investors constantly adjust their risk exposure as their mandates change.

Why Bond Prices Change Frequently (Between Major Events)

Even without a Repo rate cut or rating change, secondary market bond prices float continuously due to four dynamic drivers:

1. Expectation Shifts

Yields move **ahead** of events. If inflation data spikes on Tuesday, traders price in an RBI rate hike immediately—long before the official MPC meeting.

2. Credit Spread Fluctuations

Corporate earnings previews, sector news, or broader risk sentiment continuously expand or compress the spread over G-Secs.

3. Liquidity & Order Flow

Large institutional fund inflows/outflows, interbank liquidity shifts, and dealer inventory positioning move yields by fractions of a basis point daily.

4. Macro Data & Global Spillovers

Fluctuations in crude oil prices, FX rates (USD/INR), and US Treasury yields drive daily re-pricing across domestic markets.

- **How is the trade executed?**
- Unlike stocks on a centralized exchange, most bonds trade **Over-The-Counter (OTC)** through dealer networks.



Crucial Concept

When exiting early, Investor A does *not* get their initial investment (P_0) or the Principal back. They receive today's **Current Market Price** (P_t), which fluctuates!

- ▶ A bond's **Coupon Rate** is permanently fixed (e.g., paying 5% of face value every year).
- ▶ But because the bond's *market price* fluctuates every day, the *actual percentage return* a new buyer gets will change!

Yield to Maturity (YTM)

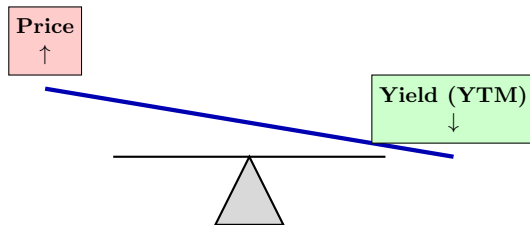
The total expected annualized return if the bond is bought at today's market price and held to maturity.

- ▶ **The Mathematical Definition:** YTM (y) is the specific internal discount rate that makes the Present Value of all remaining cash flows exactly equal to the **Current Market Price** (P):

$$P = \sum_{t=1}^T \frac{C}{(1+y)^t} + \frac{\text{Principal}}{(1+y)^T}$$

The Price-Yield Relationship

- ▶ From the pricing formula, the Yield (y) is in the denominator.
- ▶ Therefore, Bond Prices and Yields share a strict **inverse relationship**.



The Economic Intuition

If prevailing market interest rates rise, newly issued bonds will pay higher coupons. Your older bond with a lower fixed coupon becomes less attractive. To convince someone to buy it, its **price must drop** until its overall return (Yield) matches the new market environment.

Numerical Example: Selling Before Maturity

The Setup: You buy a newly issued bond today ($t = 0$).

- ▶ **Face Value (Principal):** 1,000
- ▶ **Coupon Rate:** 5% annual (Coupon = 50 per year)
- ▶ **Maturity (T):** 2 years
- ▶ **Initial Yield:** 5% $\rightarrow$ You pay exactly **1,000** (P_0).

Fast forward 1 year ($t = 1$): You collected your first 50 coupon, but now you need cash and must sell the bond. There is exactly **1 year left** to maturity.

Remaining cash flow at $t = 2$: **1,050** (50 coupon + 1,000 principal).

Scenario A: Market Rates Rise

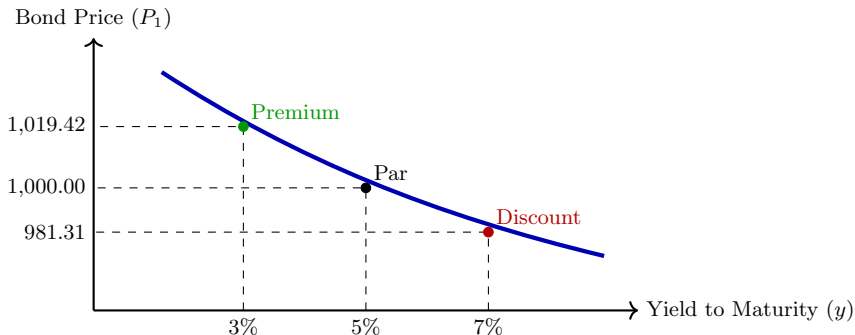
Suppose new 1-year bonds now pay **7%**.

- ▶ Buyers demand a 7% yield to buy your bond (YTM = 0.07).
- ▶ **New Market Price (P_1)** = $\frac{1050}{(1+0.07)^1} = \mathbf{981.31}$; **Discounted price**

Scenario B: Market Rates Fall

Suppose new 1-year bonds now pay only **3%**.

- Buyers only demand a 3% yield ($YTM = 0.03$).
- **New Market Price (P_1)** = $\frac{1050}{(1+0.03)^1} = 1,019.42$
- *Result:* You sell at a **Premium**. You made a capital gain!



The Asset Spectrum: From Fixed Income to Equities

- ▶ **Bonds:** Contractual, *deterministic* cash flows (Coupons + Principal).
- ▶ **Equities (Stocks):** Ownership in a company with *uncertain* future returns.

Characteristics of Equity (S)

- ▶ No guaranteed cash flows (Dividends are discretionary).
- ▶ Residual claim on company assets.
- ▶ **Spot Price (S_0)** fluctuates constantly based on market expectations.

Bond Cash Flows
Predictable / Fixed

Stock Price (S_t)
Volatile / Random



The Bridge to Derivatives

Because stock prices (S_0) move unpredictably in the spot market, investors and corporations face **Price Risk**. We need new financial contracts—**Derivatives**—to manage this volatility!

Forwards and Futures

The Problem: Price Risk in the Spot Market

- ▶ Until now, we discussed borrowing and lending cash. What if we are dealing with **assets** (stocks, commodities, currencies)?
- ▶ **Spot Market:** Buying or selling an asset *today* at today's price (S_0).
- ▶ **The Risk:** What if you *need* to buy or sell the asset in the future ($t = T$), but you don't know what the future price (S_T) will be?

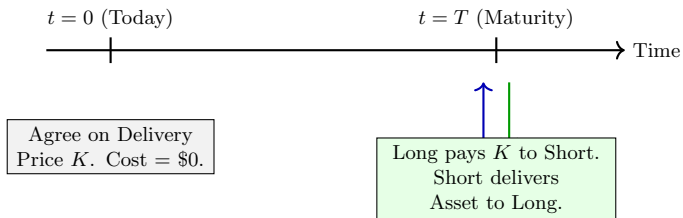


How can they eliminate this uncertainty today?

What is a Forward Contract?

A legally binding agreement between two parties to buy or sell an asset at a specified future time (T) for a price agreed upon **today** (K).

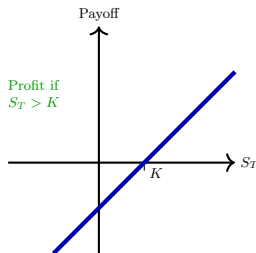
- ▶ **Long Position:** Agrees to *buy* the asset. (e.g., The Baker)
- ▶ **Short Position:** Agrees to *sell* the asset. (e.g., The Farmer)
- ▶ **Crucial Feature:** *No money changes hands today ($t = 0$).* The contract value at initiation is zero.



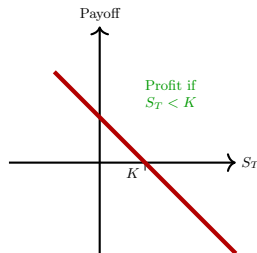
Payoff Profiles: The Linear Nature of Forwards

- ▶ At maturity T , the actual market price of the asset is S_T .
- ▶ Because they locked in price K , their profit/loss depends entirely on the difference between S_T and K .

$$\text{Long Payoff} = S_T - K$$



$$\text{Short Payoff} = K - S_T$$



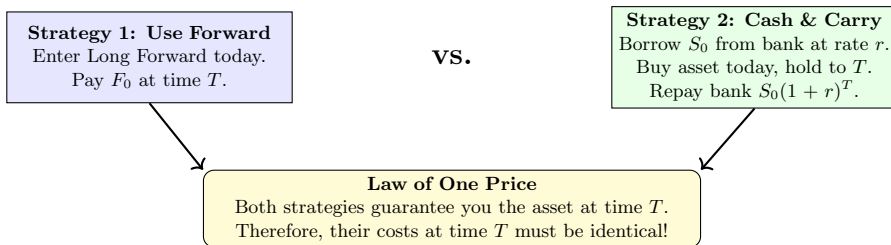
Key Observation

This payoff is perfectly **linear** and symmetric. One party's gain is exactly the other party's loss.

Pricing a Forward: The Arbitrage Argument

- ▶ How do we determine the fair delivery price (F_0) today so that the initial value of the contract is exactly \$0?
- ▶ **Do we need to guess where the stock price S_T is going? No!**
- ▶ We use **Static Replication** (Cash-and-Carry Arbitrage).

Suppose you want to own the asset at time T . You have two choices today ($t = 0$):



The Fundamental Forward Equation

Equating the costs of Strategy 1 and Strategy 2 at maturity T :

$$F_0 = S_0(1 + r)^T$$

- ▶ F_0 = Fair Forward Price
- ▶ S_0 = Current Spot Price of the underlying asset
- ▶ r = Risk-free interest rate
- ▶ T = Time to maturity

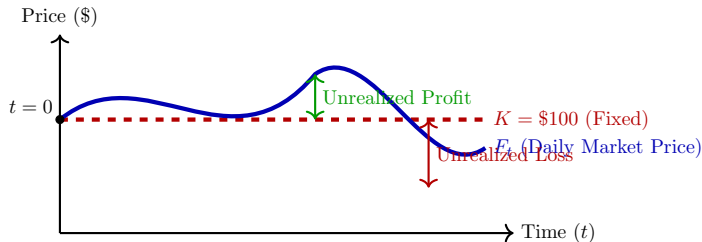
The Crucial Paradigm Shift

Notice what is **missing** from this formula. We priced a derivative without ever calculating the expected future price of the stock, probabilities, or risk aversion!

Because the payoff is *linear*, we can price it perfectly using just the **spot price** and the **bond (borrowing) rate**.

Locked Contract vs. Floating Market: K vs. F_t

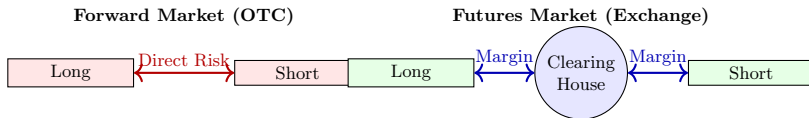
- ▶ **The Strike Price (K):** Written on the paper contract at $t = 0$. It is **fixed forever**.
- ▶ **The Market Forward Price (F_t):** Fluctuates every single day based on changes in spot price S_t and interest rates.



Why Margins Exist

The clearinghouse measures the gap ($F_t - K$) every day. To ensure the party suffering a loss can cover their obligation, collateral (**Margin**) is collected continuously based on these movements.

- ▶ **The Flaw in Forwards:** They are private, Over-The-Counter (OTC) contracts. If S_T crashes, the Long position might default and run away. This is **Counterparty Credit Risk**.
- ▶ **The Solution: Futures Contracts.**
 - ★ Standardized and traded on an Exchange.
 - ★ **Clearinghouse:** Steps in as the middleman (Buyer to every Seller, Seller to every Buyer).
 - ★ **Margin & Mark-to-Market:** Profits and losses are settled *daily*, preventing massive debts from building up at maturity.



- ▶ Instead of waiting until maturity, losses and gains are settled **every afternoon** (Mark-to-Market).
- ▶ **Key Terminology:**
 - ★ **Initial Margin:** Mandatory cash deposit required to open the position.
 - ★ **Maintenance Margin:** The floor level your account balance is allowed to reach before action is required.
 - ★ **Variation Margin:** The required cash top-up when the balance falls below the maintenance floor.

Numerical Example Setup

- ▶ **Position:** Long 1 Futures Contract = 100 shares of Stock XYZ.
- ▶ **Day 0 Futures Price (F_0):** \$100 per share (Total Notional Value = \$10,000).
- ▶ **Initial Margin Deposit:** \$1,000 (10% of contract value).
- ▶ **Maintenance Margin Floor:** \$700 (70% of Initial Margin).

Margin Account Ledger: A Step-by-Step Example

Setup: Long 100 shares at $F_0 = \$100$. Initial Margin = \$1,000. Maintenance Floor = \$700.

Day	Futures (F_t)	Daily P/L	Account Balance	Action / Call
0	\$100	–	\$1,000	Deposit \$1,000
1	\$103	+\$300	\$1,300	None (Can withdraw \$300)
2	\$95	–\$800	\$500	MARGIN CALL! (< \$700)
2 (End)	\$95	–	\$1,000	Deposit \$500 (Variation)
3	\$97	+\$200	\$1,200	None

Crucial Margin Call Rule

When hit with a Margin Call, you must deposit enough cash to restore the balance back to the **Initial Margin (\$1,000)**, NOT just the Maintenance Floor (\$700)!

Options

- ▶ With Forwards and Futures, both parties are **obligated** to transact at maturity T .
- ▶ What if you only want to transact if the market moves in your favor?

What is an Option?

A contract that gives the buyer the **right, but not the obligation**, to buy or sell an underlying asset at a specified strike price (K) on or before a specified date (T).

Call Option

- ▶ The right to **BUY**.
- ▶ You want the asset price (S_T) to go **UP**.

Put Option

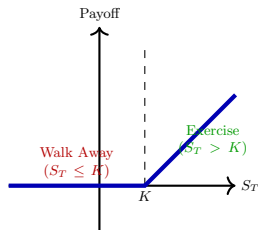
- ▶ The right to **SELL**.
- ▶ You want the asset price (S_T) to go **DOWN**.

The Catch: The Premium

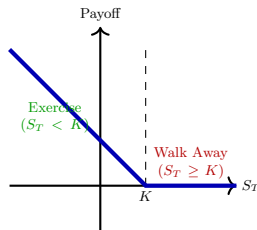
Because the buyer has a choice (and the seller has all the risk), entering an option contract is **not free**. The buyer must pay a price upfront at $t = 0$, called the **Option Premium**. Our goal is to figure out how to calculate this premium!

- At maturity T , the option buyer will only exercise the contract if it is profitable. Otherwise, they walk away (payoff is zero).

$$\text{Long Call Payoff} = \max(S_T - K, 0)$$



$$\text{Long Put Payoff} = \max(K - S_T, 0)$$

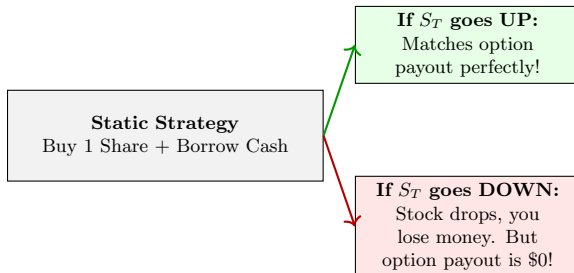


The Asymmetry

Unlike Forwards, the payoff is **kinked** at the strike price K . The downside is strictly limited to zero, while the upside can be massive.

The Breakdown of Static Replication

- **Recall:** We priced Forwards easily using a “Cash and Carry” strategy (borrow money, buy 1 share of stock today, hold until T).
- **Can we do this for a Call Option?**



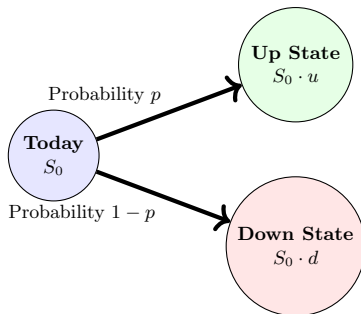
The Pricing Dilemma

You **cannot** statically mix a straight line (stock) and a flat line (bond) to perfectly match a kinked curve (option) across all possible futures. The static arbitrage argument fails!

The Hook: Enter Discrete State Models

- ▶ If static replication fails, how do we price this non-linear payoff today?
- ▶ We can't just take a simple average of future prices, because risk aversion matters!

The Insight: We need to model the *possible states* the stock price can take in the future, and build a portfolio that adjusts to those states.



The 1-Period Binomial Model
Instead of an infinite continuous universe, what if the stock can only do two things tomorrow?